Causal effect of PDCD1 protein on type 1 diabetes: MR-Hevo analysis

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Contents

1 Introduction

We estimate the causal effect of circulating PDCD1 (programmed cell death protein 1) levels on risk of type 1 diabetes (T1D) using Mendelian randomization with the regularized horseshoe prior (MR-Hevo).

Results are presented from two analyses:

1. **Individual-level T1D outcome (Zhou et al.)**: instrument-exposure coefficients from the UK Biobank Olink proteomics study ($N = 33{,}902$); instrument-outcome coefficients estimated from individual-level genotype–phenotype data.
2. **Summary-level T1D outcome, UK Biobank SNP–protein coefficients**: both sets of coefficients from summary statistics, with SNP–protein associations from the UK Biobank Olink study ($N = 33{,}902$) and SNP–T1D associations from Iakovliev et al. (Am J Hum Genet 2023; 4,964 cases, 7,497 controls). LD matrices from the UK Biobank EUR panel (362,063 samples, 18 million variants) were used to construct scalar instruments.

Scalar genetic instruments are constructed from loci associated with PDCD1 at $p < 10^{-6}$, excluding the HLA region (chromosome 6: 25–34 Mb). Each scalar instrument Z is a linear combination of SNPs within the locus, constructed so that $\mathrm{SD}(Z) = 1$ in the target dataset. The instrument-exposure coefficient $\hat{\alpha}$ is therefore the expected change in the exposure per change of 1 SD in the scalar instrument Z; it can be interpreted directly as the strength of the instrument. Throughout this report all displayed values of $\hat{\alpha}$ and $\hat{\gamma}$ are in per-SD(Z) units, obtained by multiplying the per-unit-Z coefficients (as stored in the intermediate `.rds`/`.csv` files) by SD(Z).

2 Data sources

The following source data files are used in this analysis.

File	Description	Source
PDCD1_Q15116_OID21396_v1_Oncology.tar	SNP–protein GWAS summary statistics for PDCD1 (UniProt Q15116; Olink Oncology panel, assay OID21396); $N = 33{,}902$ UK Biobank participants of European ancestry	UK Biobank Pharma Proteomics Project (UKB-PPP); Sun et al. (2023) *Nature* 622:329–338
gwasresults.RData.gz	Type 1 diabetes GWAS summary statistics; 4,964 cases and 7,497 controls	Iakovliev et al. (2023) *Am J Hum Genet* 110:1718–1732
pdcd1.mr.coeffs.csv	Scalar instrument–exposure ($\hat{\alpha}$) and instrument–outcome ($\hat{\gamma}$) coefficients from the individual-level MR analysis, with scalar instruments constructed using LD from the 1000 Genomes EUR panel	Computed by Zhou et al. from UK Biobank Olink proteomics and individual-level T1D genotype–phenotype data
LD_Eur_18mvariants/int8/ (genoscores)	Block-LD matrix for UK Biobank participants of European ancestry; $N = 362{,}063$ samples; $\sim$18 million variants; GRCh37 coordinates; quantized to int8	UK Biobank; computed using magenpy
refpop/kg.2020.hg38.eur.bim	Variant index (rsid, chromosome, position hg38) for the 1000 Genomes EUR reference panel; 503 samples; used for (chr, pos) $\rightarrow$ rsid lookup	1000 Genomes Project phase 3, GRCh38 liftover
refGene.txt.gz	RefSeq protein-coding gene annotations (NM_ accessions); used to label loci with the nearest gene	UCSC Genome Browser, hg38 assembly

3 Individual-level T1D outcome (Zhou et al.)

38 scalar instruments from the UK Biobank Olink proteomics study ($N = 33{,}902$), with instrument-outcome coefficients estimated from individual-level genotype-outcome data in the T1D case-control study (Zhou et al.). Nearest Gene lists all protein-coding genes at the locus.

Table 2: Instrument coefficients — UK Biobank, individual-level T1D outcome (Zhou et al.). Nearest Gene: protein-coding gene(s) nearest to or overlapping the locus. Num SNPs: SNPs in locus; $\hat{\alpha}$ and $\hat{\gamma}$: instrument-exposure and instrument-outcome coefficients per SD of the scalar instrument (SE in parentheses). *: $|\hat{\gamma}/\mathrm{SE}(\hat{\gamma})| > 2.58$ (two-sided $p < 0.01$).

Nearest Gene	Chr	Pos (Mb)	Width (Mb)	Num SNPs	$\hat{\alpha}$ (SE)	$\hat{\gamma}$ (SE)	*
ACAP3	1	1.21	0.16	99	0.0253 (0.0051)	0.0188 (0.0206)	
ARTN	1	43.48	0.64	420	0.0540 (0.0051)	0.0220 (0.0201)	
PTPN22	1	113.53	0.60	234	0.0293 (0.0051)	0.1635 (0.0197)	*
HDGF	1	156.77	0.07	71	0.0289 (0.0051)	0.0127 (0.0202)	
AXDND1	1	179.31	0.49	30	0.0357 (0.0051)	-0.0201 (0.0204)	
NEK7	1	198.30	0.16	33	0.0252 (0.0051)	-0.0191 (0.0202)	
C1orf147	1	206.50	0.02	2	0.0257 (0.0051)	-0.0087 (0.0201)	
RN7SL51P	2	62.26	0.12	61	0.0406 (0.0051)	0.0181 (0.0207)	
GLS	2	190.64	0.47	59	0.0363 (0.0051)	0.0262 (0.0199)	
CTLA4	2	203.83	0.10	129	0.0400 (0.0051)	0.1254 (0.0202)	*
PDCD1	2	241.00	0.23	24	0.1237 (0.0051)	-0.0130 (0.0203)	
ARHGEF3	3	56.90	0.02	2	0.0266 (0.0051)	-0.0194 (0.0203)	
GPR160	3	170.00	0.20	132	0.0339 (0.0051)	0.0134 (0.0203)	
ST6GAL1	3	187.02	0.00	6	0.0266 (0.0051)	-0.0325 (0.0201)	
CAPSL	5	35.80	0.12	42	0.0310 (0.0051)	0.0394 (0.0203)	
CDC25C	5	138.25	0.04	3	0.0255 (0.0051)	-0.0071 (0.0199)	
	6	51.28	0.02	9	0.0327 (0.0051)	-0.0414 (0.0202)	
BACH2	6	90.27	0.03	2	0.0265 (0.0051)	0.1428 (0.0199)	*
TAGAP-AS1	6	159.05	0.04	20	0.0271 (0.0051)	0.0617 (0.0203)	*
CD274	9	5.48	0.02	25	0.0277 (0.0051)	0.0250 (0.0200)	
B3GALT9	9	120.75	0.22	138	0.0308 (0.0051)	0.0589 (0.0203)	*
IL15RA	10	5.69	0.38	63	0.0391 (0.0051)	0.0715 (0.0201)	*
ATP5MC1P7	10	69.38	0.16	183	0.0538 (0.0051)	0.0039 (0.0202)	
DYDC1	10	80.28	0.24	225	0.0291 (0.0051)	-0.0178 (0.0201)	
SH2B3	12	109.92	2.88	1186	0.0714 (0.0051)	0.1157 (0.0203)	*
ACADS	12	120.70	0.25	249	0.0328 (0.0051)	0.0239 (0.0200)	
ALDH1A2	15	58.46	0.04	10	0.0402 (0.0051)	0.0124 (0.0200)	
DRAIC	15	69.69	0.07	39	0.0285 (0.0051)	0.0465 (0.0201)	
PEAK1	15	77.15	0.11	6	0.0248 (0.0051)	0.0249 (0.0201)	
CLEC16A	16	11.02	0.09	26	0.0260 (0.0051)	0.1642 (0.0207)	*
ACADVL	17	7.16	0.20	38	0.0473 (0.0051)	0.0288 (0.0200)	
FLJ40194	17	49.24	0.16	106	0.0348 (0.0051)	0.0255 (0.0202)	
CEP131	17	81.17	0.13	147	0.0468 (0.0051)	0.0205 (0.0201)	
DAPK3	19	3.95	0.16	73	0.0371 (0.0051)	0.0232 (0.0202)	
AP1M2	19	10.53	0.55	275	0.0481 (0.0051)	-0.0199 (0.0203)	
FUT2	19	48.65	0.10	105	0.0321 (0.0051)	0.1395 (0.0202)	*
AIRE	21	44.29	0.00	5	0.0256 (0.0051)	-0.0022 (0.0201)	
OGFRP1	22	42.19	0.10	28	0.0255 (0.0051)	-0.0181 (0.0202)	

4 UK Biobank — summary-level T1D outcome

39 scalar instruments from the UK Biobank Olink proteomics study ($N = 33{,}902$). Instrument-outcome coefficients from Iakovliev et al.

Table 3: Instrument coefficients — UK Biobank SNP–protein coefficients, summary-level T1D. Nearest Gene: protein-coding gene(s) nearest to or overlapping the locus. Num SNPs: SNPs in locus; eff rank: effective rank of LD matrix. SD(Z): standard deviation of scalar instrument in reference panel. $\hat{\alpha}$ and $\hat{\gamma}$: instrument-exposure and instrument-outcome coefficients per SD of the scalar instrument (SE in parentheses). *: $|\hat{\gamma}/\mathrm{SE}(\hat{\gamma})| > 2.58$ (two-sided $p < 0.01$).

Nearest Gene	Chr	Pos (Mb)	Width (Mb)	Num SNPs	eff rank	SD(Z)	$\hat{\alpha}$ (SE)	$\hat{\gamma}$ (SE)	*
SDF4	1	1.21	0.04	84	2	2.775	0.0260 (0.0051)	0.0196 (0.0183)	
ST3GAL3	1	43.33	0.79	353	9	3.149	0.0507 (0.0051)	0.0231 (0.0183)	
PTPN22	1	113.53	0.64	191	8	0.580	0.0493 (0.0051)	-0.0063 (0.0073)	
PRCC	1	156.77	0.07	56	6	0.690	0.0322 (0.0051)	0.0006 (0.0036)	
TDRD5	1	179.31	0.49	26	11	0.360	0.0411 (0.0052)	-0.0003 (0.0183)	
NEK7	1	198.30	0.16	29	2	0.878	0.0280 (0.0052)	-0.0124 (0.0183)	
IKBKE	1	206.50	0.00	1	1	0.593	0.0256 (0.0051)	-0.0087 (0.0183)	
B3GNT2	2	62.26	0.12	45	11	0.460	0.0444 (0.0053)	-0.0114 (0.0183)	
NAB1	2	190.64	0.47	45	7	0.653	0.0461 (0.0052)	0.0135 (0.0183)	
CTLA4	2	203.83	0.10	102	7	1.218	0.0524 (0.0052)	-0.0016 (0.0092)	
PDCD1	2	241.04	0.20	22	16	0.175	0.0783 (0.0053)	-0.0081 (0.0085)	
ARHGEF3	3	56.90	0.02	2	2	0.579	0.0316 (0.0052)	-0.0194 (0.0183)	
PHC3, GPR160	3	170.00	0.20	112	6	1.980	0.0396 (0.0052)	-0.0064 (0.0183)	
ST6GAL1	3	187.02	0.00	6	3	0.836	0.0279 (0.0052)	-0.0499 (0.0183)	*
IL7R	5	35.80	0.12	35	5	2.088	0.0326 (0.0052)	0.0261 (0.0183)	
GFRA3	5	138.16	0.09	5	4	0.262	0.0269 (0.0052)	-0.0092 (0.0169)	
PKHD1	6	51.28	0.02	5	3	1.201	0.0269 (0.0052)	-0.0333 (0.0183)	
BACH2	6	90.27	0.03	2	2	0.494	0.0269 (0.0051)	0.1289 (0.0183)	*
	6	159.05	0.04	20	5	0.754	0.0287 (0.0051)	-0.0338 (0.0183)	
TSPAN33	7	129.10	0.04	4	2	0.978	0.0286 (0.0051)	0.0127 (0.0183)	
CD274	9	5.48	0.02	16	2	1.738	0.0282 (0.0051)	0.0240 (0.0183)	
PHF19	9	120.75	0.22	108	3	4.386	0.0305 (0.0051)	-0.0533 (0.0183)	*
IL15RA	10	5.69	0.38	46	5	0.906	0.0402 (0.0051)	0.0692 (0.0183)	*
TACR2	10	69.38	0.16	148	15	1.376	0.0602 (0.0052)	0.0015 (0.0183)	
TSPAN14	10	80.28	0.24	180	6	2.740	0.0314 (0.0051)	0.0249 (0.0183)	
SH2B3	12	110.00	2.79	995	15	3.064	0.0736 (0.0051)	0.0009 (0.0181)	
SPPL3	12	120.75	0.19	206	6	2.523	0.0374 (0.0051)	0.0026 (0.0184)	
LIPC	15	58.46	0.06	16	7	0.670	0.0507 (0.0052)	-0.0137 (0.0299)	
RPLP1	15	69.69	0.07	32	4	1.216	0.0307 (0.0051)	0.0101 (0.0183)	
PEAK1	15	77.15	0.09	5	4	0.562	0.0306 (0.0052)	0.0302 (0.0183)	
CLEC16A	16	11.04	0.08	19	5	1.346	0.0273 (0.0051)	0.1101 (0.0183)	*
ASGR1	17	7.06	0.29	40	22	0.182	0.0881 (0.0052)	-0.0022 (0.0051)	
ZNF652	17	48.49	0.91	109	9	1.361	0.0438 (0.0051)	0.0225 (0.0179)	
SLC38A10	17	81.17	0.13	127	10	1.821	0.0471 (0.0051)	-0.0016 (0.0187)	
PIAS4	19	3.95	0.14	59	13	0.765	0.0428 (0.0052)	0.0145 (0.0183)	
DNM2, SMARCA4, KRI1, SLC44A2, C19orf38, CDKN2D, ATG4D, AP1M2, ILF3, YIPF2, CARM1, QTRT1, TMED1, TIMM29	19	10.53	0.55	221	12	2.066	0.0516 (0.0051)	0.0121 (0.0182)	
FUT2	19	48.65	0.10	93	5	3.682	0.0324 (0.0052)	0.0762 (0.0183)	*
AIRE	21	44.29	0.00	3	1	0.764	0.0263 (0.0051)	0.0033 (0.0183)	
TCF20	22	42.19	0.10	19	3	0.877	0.0296 (0.0051)	0.0066 (0.0183)	

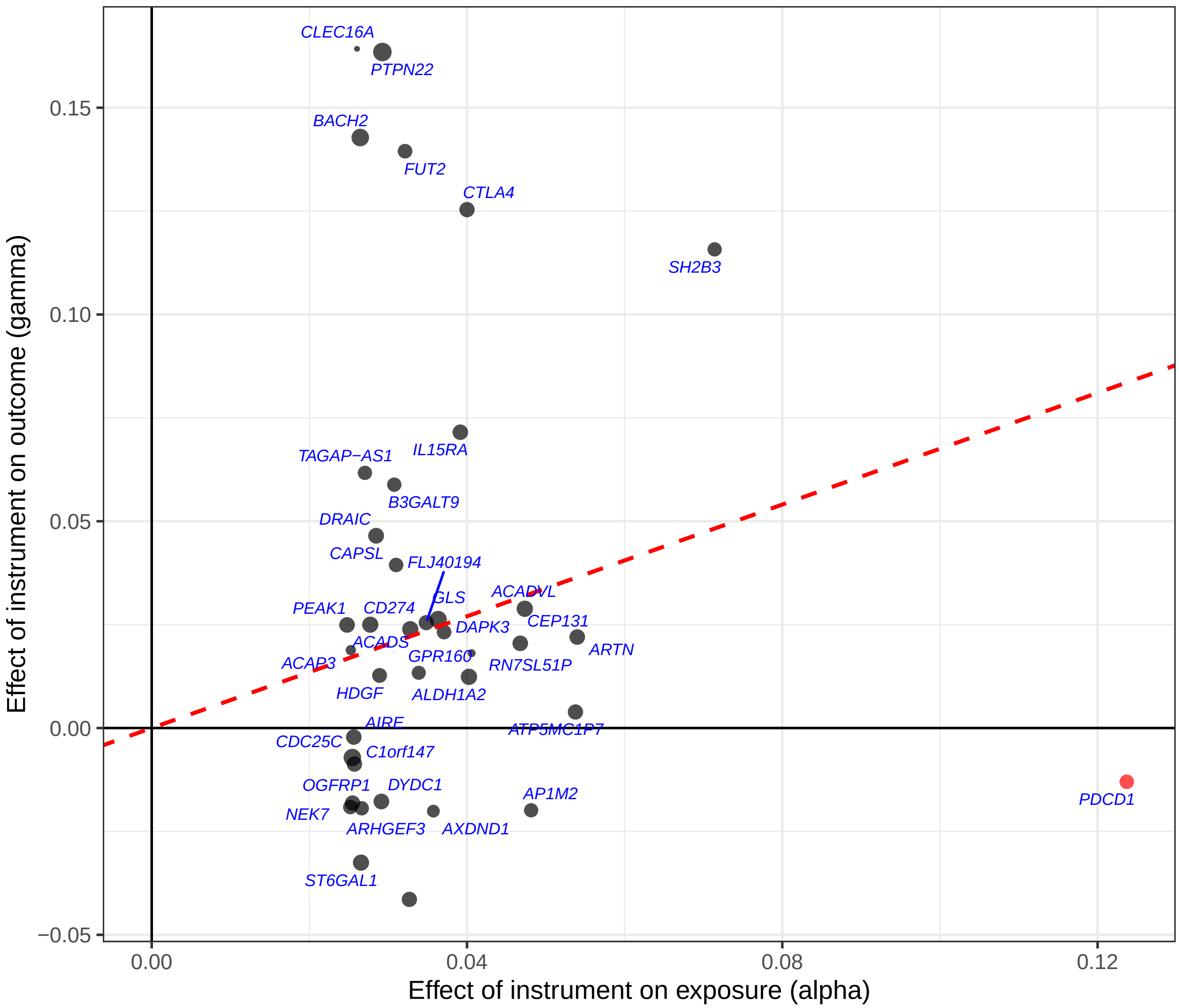

Figure 1: UK Biobank, individual-level T1D (Zhou et al.): $\hat{\gamma}$ vs $\hat{\alpha}$ (both per SD of scalar instrument) for 38 instruments. IVW slope estimated from trans-pQTLs only: $\hat{\theta} = 0.675$ (SE 0.098, $p = 5 \times 10^{-10}$). Red dashed line: IVW slope (trans only). Red point: PDCD1 cis-pQTL.

5 Comparison of $\hat{\alpha}$ coefficients across analyses

6 Comparison of $\hat{\gamma}$ coefficients across analyses

MR-Hevo posterior results not yet available. Run `run_mrhevo_pdcd1_t1d.R` *to generate* `mrhevo_pdcd1_t1d_fit.rds`.

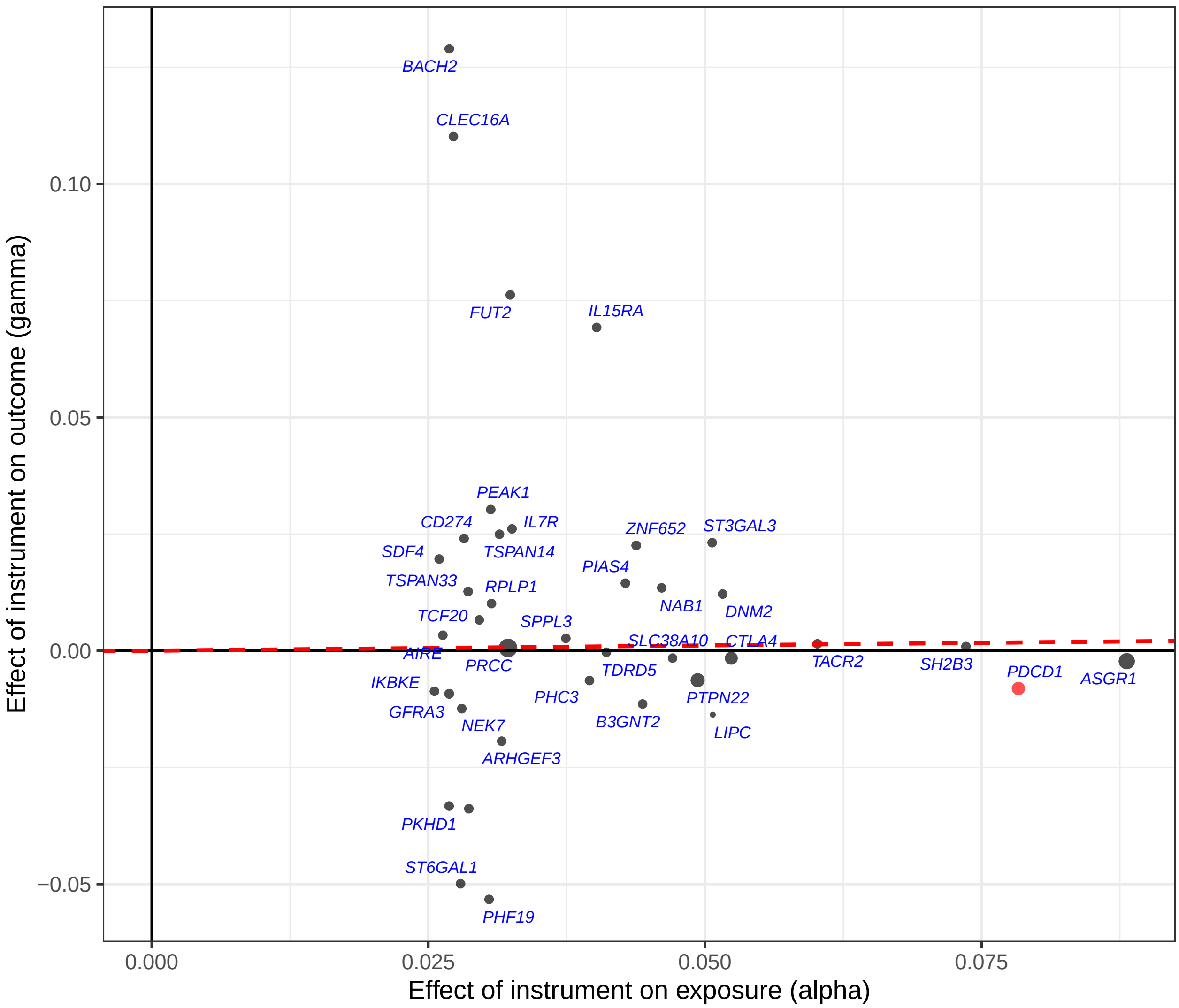

Figure 2: UK Biobank (summary-level T1D): $\hat{\gamma}$ vs $\hat{\alpha}$ (both per SD of scalar instrument) for 39 instruments. IVW slope estimated from trans-pQTLs only: $\hat{\theta} = 0.022$ (SE 0.041, $p = 0.6$). Red dashed line: IVW slope (trans only). Red point: PDCD1 cis-pQTL.

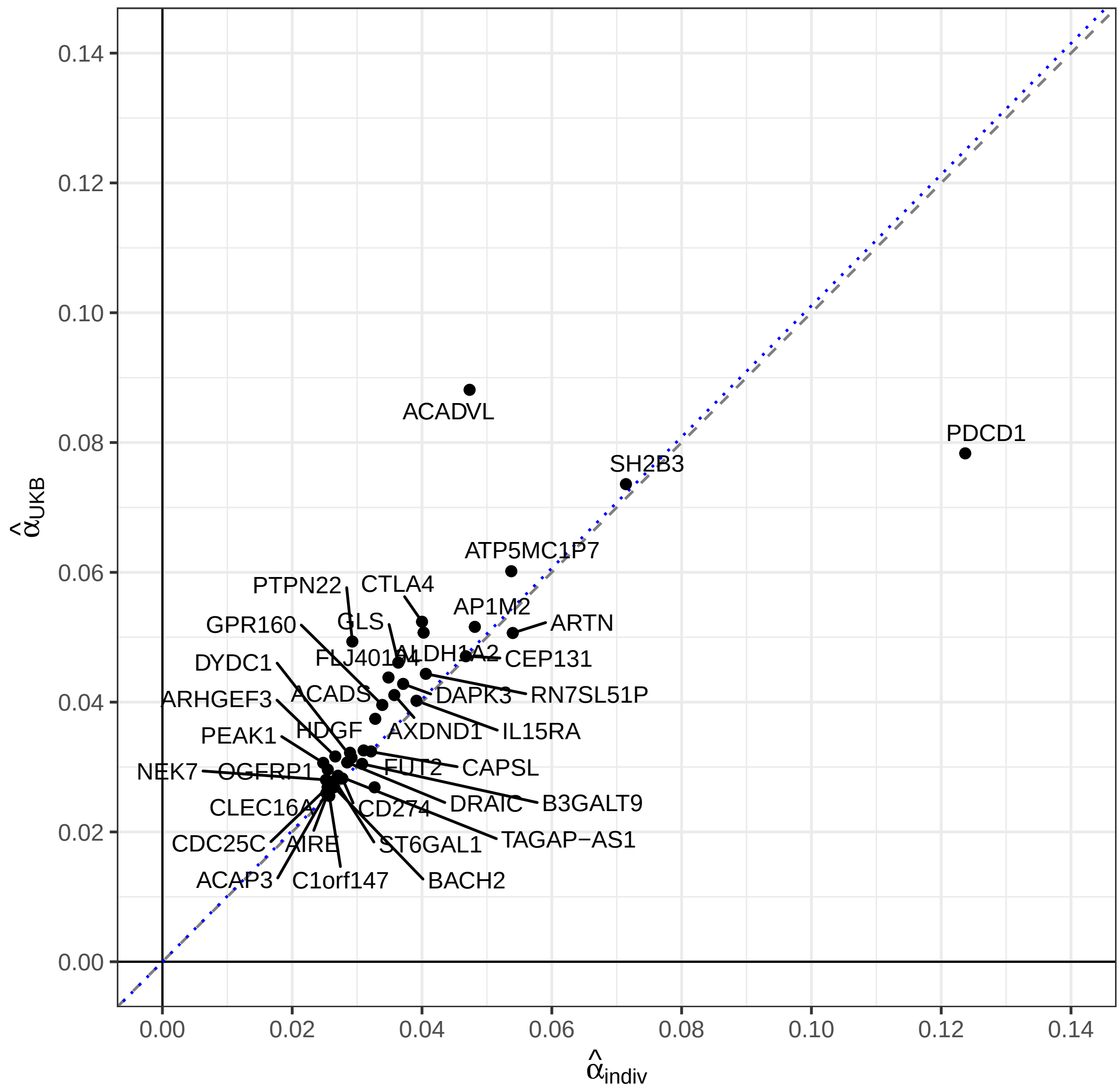

Figure 3: Instrument-exposure coefficient $\hat{\alpha}$ (per SD of scalar instrument) from the individual-level analysis (x-axis) versus the UK Biobank summary-level analysis (y-axis). Loci matched by chromosomal overlap. Grey dashed line: identity ($y = x$). Blue dotted line: OLS regression through the origin.

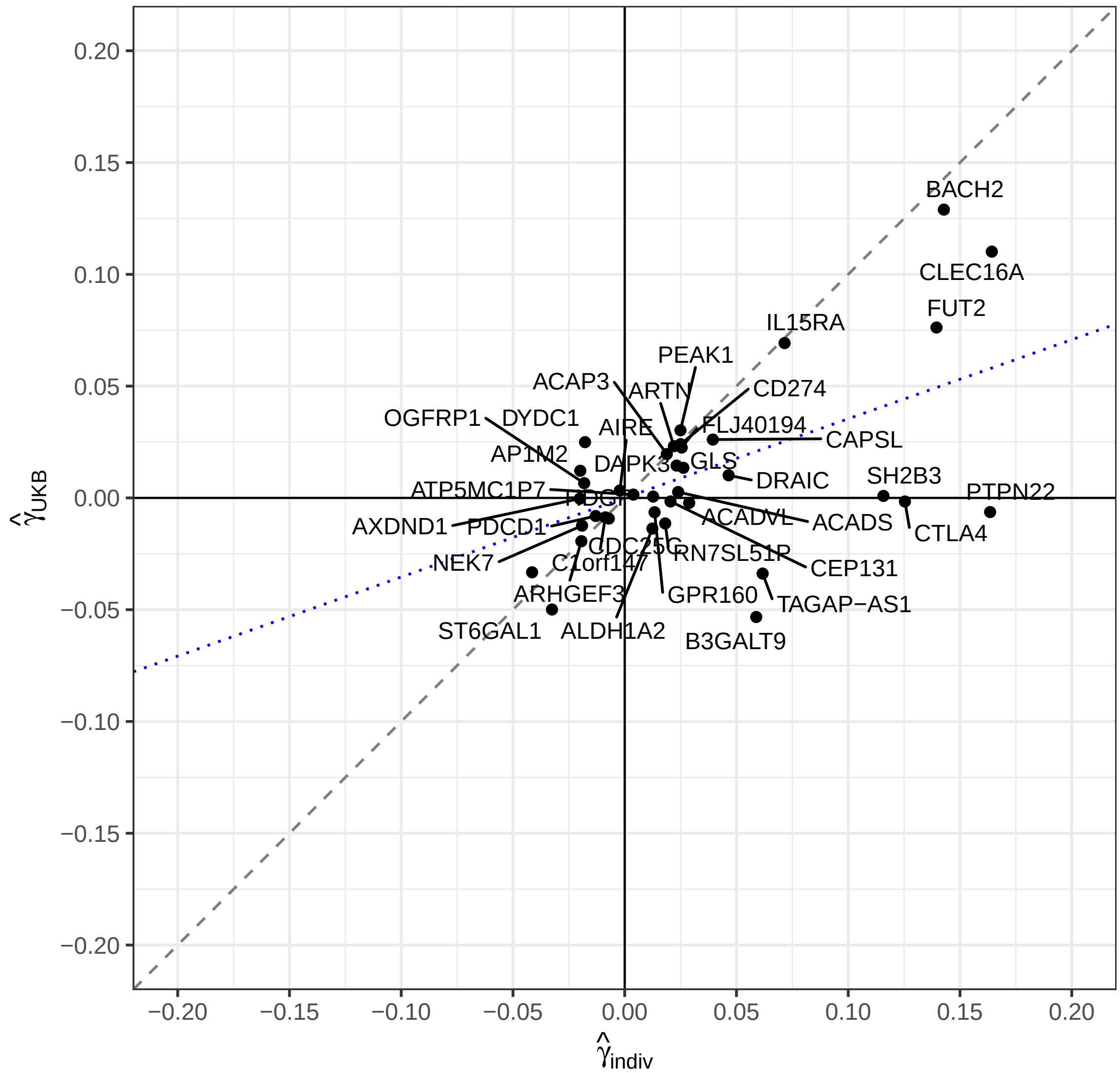

Figure 4: Instrument-outcome coefficient $\hat{\gamma}$ (per SD of scalar instrument, in log-odds units) from the individual-level analysis (x-axis) versus the UK Biobank summary-level analysis (y-axis). Loci matched by chromosomal overlap. Grey dashed line: identity ($y = x$). Blue dotted line: OLS regression through the origin.